2. **Say True or False.**

(i) 8, 15, 17 is a Pythagorean triplet.

(ii) In a right angled triangle, the hypotenuse is the greatest side.

(iii) In any triangle the centroid and the incentre are located inside the triangle.

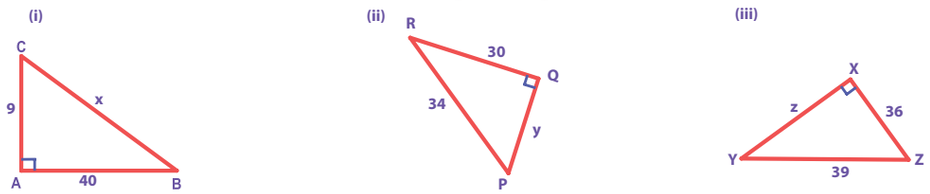
(iv) The centroid, orthocentre, and incentre of a triangle are collinear.

(v) The incentre is equidistant from all the vertices of a triangle.

3. Check whether given sides are the sides of right-angled triangles, using Pythagoras theorem.

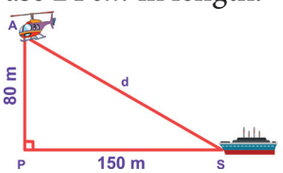
(i) 8,15,17 (ii) 12,13,15 (iii) 30,40,50 (iv) 9,40,41 (v) 24,45,51

4. Find the unknown side in the following triangles.

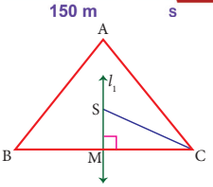


5. An isosceles triangle has equal sides each 13 *cm* and a base 24 *cm* in length. Find its height.

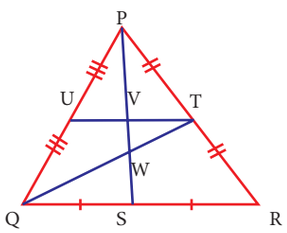
6. Find the distance between the helicopter and the ship.



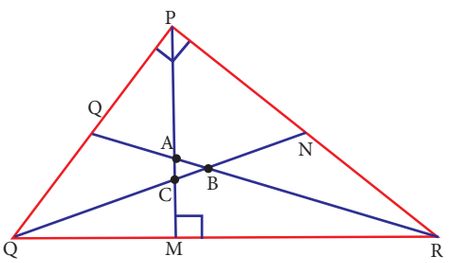
7. In triangle ABC, line l_1 is a perpendicular bisector of BC. If BC=12 cm, SM=8 cm, find CS.



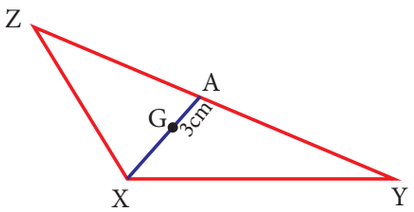
8. Identify the centroid of ΔPQR.



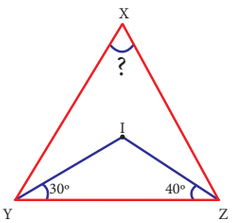
9. Name the orthocentre of ΔPQR.



10. In the given figure, A is the midpoint of YZ and G is the centroid of the triangle XYZ. If the length of GA is 3 cm, find XA.



11. If I is the incentre of ΔXYZ, $\angle IYZ = 30°$ and $\angle IZY = 40°$, find $\angle YXZ$.



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